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Subneting In CISCO PACKET TRACER Simulator

AIM:

Implementation of subneting in CISCO PACKET TRACER Simulator.

1) Classless IP subneting is a technique that allows for more difference use of IP addresses by allowing for subset masks that are not just the default masks for each IP class.

2) CREATING A NETWORK TOPOLOGY:

The first step in implementing Classless IP subnetting is to create a network topology in packet Tracer. To create a network topology in packet Tracer, Select the "NEW" a blank network topology that we can use to add devices.

3) ADDING THE DEVICES:

Once we have created our network topology, we can add devices to it, Here, we will be adding routers, witches and PCs. To add a devices, select the device from the bottom left corner and drag it onto the network topology.

A) Configure Router R1

*open R1 → CLI

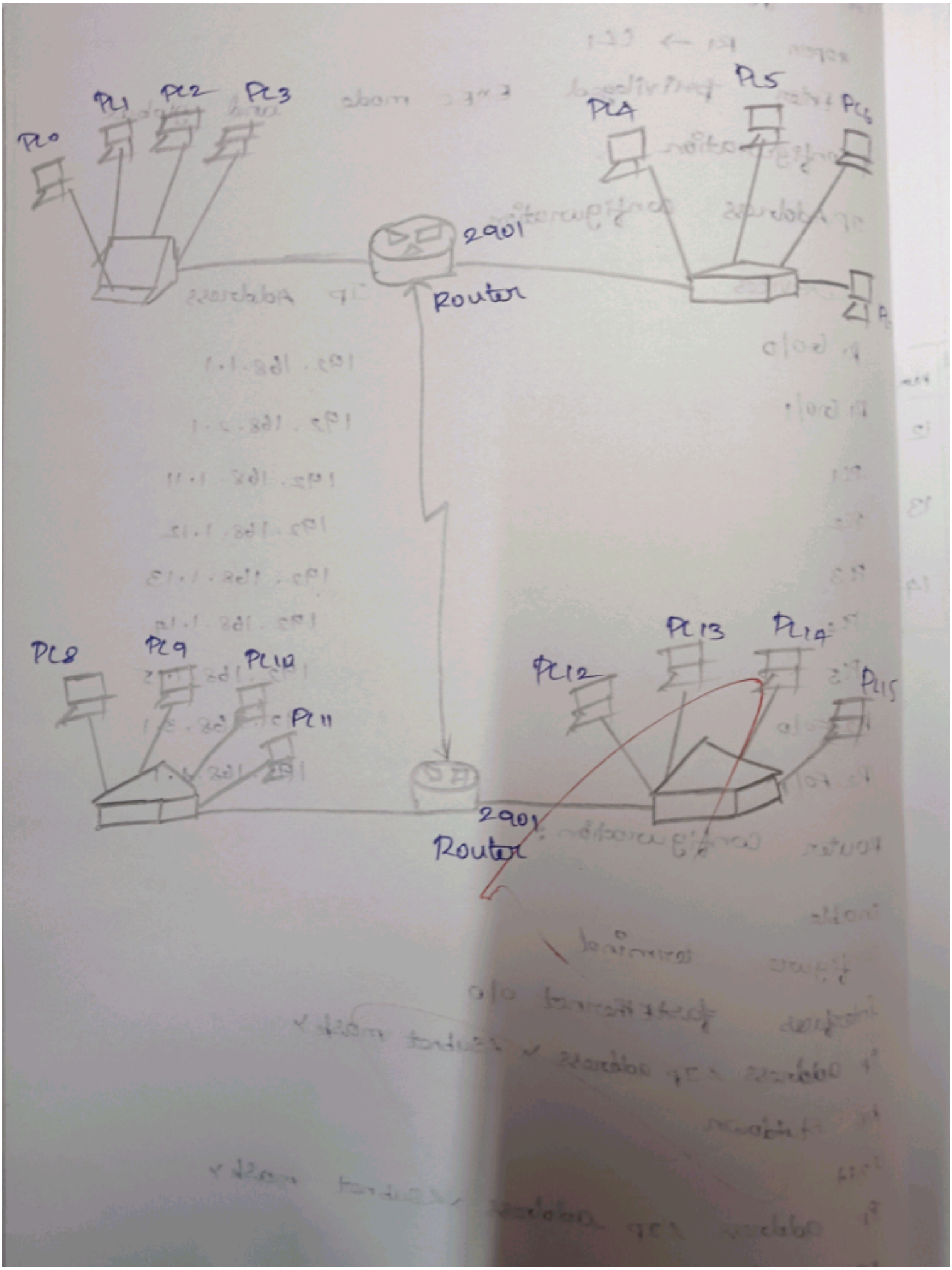
*Enter privilaged EXEC mode and global Configuration

IP Address configuration

Devices	IP Address
R1 G0/0	192.168.1.1
R1 G0/1	192.168.2.1
PC1	192.168.1.11
PC2	192.168.1.12
PC3	192.168.1.13
PC4	192.168.1.14
PC5	192.168.1.15
R2 Fo/0	192.168.3.1
R2 Fo/1	192.168.4.1

Router Configuration:

enable
configure terminal
interfaces fastEthernet 0/0
ip address <IP address> <subnet mask>
no shutdown
exit
Ip address <IP address> <Subnet mask>
no studown
exit
Switch configuration
enable
Configure terminal
interface FastEthernal 0/1


PC0
PC1
PC2
PC3
PC4
PC5
PC6
2901
Router
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
2901
Router

PC configuration:

Assign the required IP address, subnet mask, default gateway and DNS server to each PC.

Testing:

Use the Ping command from the pc command prompt to

6. configure the switch: *Right click switch CLI

enable
configure terminal
interface FastEthernet 0/1
switchport mode access
exit
interface FastEthernet 0/2
switchport mode access
exit

*these configure two switch ports in access mode for connecting PCs.

7. configure the GigabitEthernet interface: Right click Router → CLI and enter:

enable
configure terminal
interface GigabitEthernet 0/0
ip address {IP address} {subnet mask}
no shutdown
exit

Testing the network

Open a command prompt on each PC and try to ping the other PC.

Fire	Last Status	Source	Destination	Type	Color	Time (sec)	Periodic	N
0	Successful	PC4(2)	Router2	ICMP		0.000	N	
0	Successful	PC4(2)	PC2(1)(1)	ICMP		0.000	N	
0	Successful	PC0	Router0	ICMP		0.000	N	

between the devices.

* open the connect prompt on each PC
* Use the ping command to ping another PC.
* If the ping is successful, the network is working properly.
* We can also use ping to test connecting between the router and PCs.

RESULT:

The network was successfully configured using Classless IP subnetting between the router and PCs was verified using the ping command.